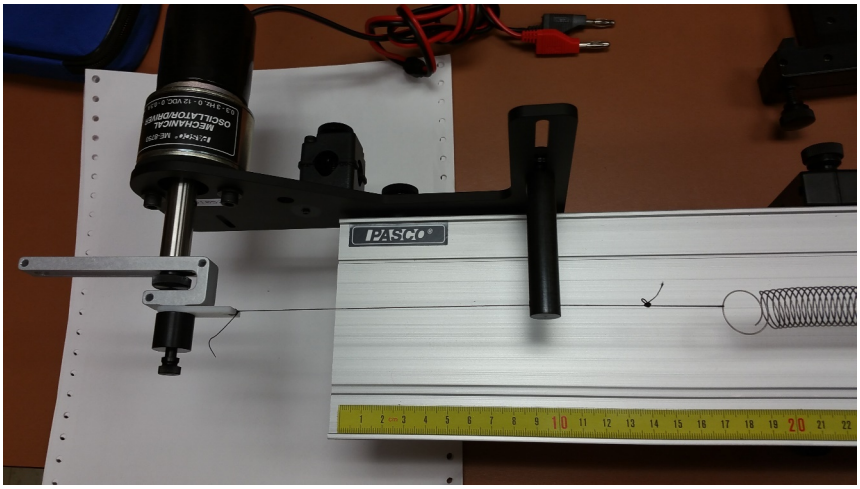
Lab 5 | Forced vibration

Theory: November 9. Laboratory: your section, November 17-20, 2026. Complete Pre-Lab 5 before your section. Check Canvas for changes.

Why this test matters

Repeated motion in motors, pumps and positioning machines can excite a structural resonance. Measure how a spring-mounted cart responds to a short impact and to periodic motion; compare the response with a one-degree-of-freedom model.



Actual apparatus: motor-driven spring end and track from the prior course handout. The motor moves the spring end; the cart sensor measures cart position.

1. Measure before recording

Record the bare cart mass, every added hex rod, and all mounted damper hardware. Use the total moving mass m [kg] for each configuration. Label the two springs and record their stiffnesses: fixed-side k_1 and driven-side k_2 [N/m].

Use your Lab 4 spring calibration if these are the same springs. Otherwise hang a known total mass, record force versus time with the Vernier force sensor, and fit A sin(Bt + phase) + offset. B is in rad/s, so $k \approx m_{hanging}B^2$ for light damping. If the fit reports f [Hz], use B = 2πf. Include the hanger mass. Use the separate Spring stiffness package on the Lab 4 page.

2. Connect Vernier Graphical Analysis

Power on the cart. In Graphical Analysis choose Sensor Data Collection, select the serial number on your cart, then Done. Select Position versus Time, position in meters, and a collection rate of 25 Hz or less as used by the laboratory computers. Level the track; keep both springs stretched near their working length (about 15 cm) without overstretching.

With the cart at equilibrium, use the sensor menu to Zero Position. Click Collect before applying the input. After each run choose Stop, then Export Data → CSV. Reopen one export and confirm time [s] and position [m]. Save one run per file with a descriptive name. Keep raw files unchanged.

3. Three impulse responses

Remove magnetic dampers. Attach a spring on each side, with both outer ends stationary. Start collection, let the cart rest at equilibrium, give its body one brief gentle horizontal tap with your fingers, and remove your hand immediately. Do not push continuously or strike the end stop. Record several clean cycles until the motion is small.

Record light, medium and slightly stronger gentle taps in the same direction, starting from rest each time with the same mass and springs. Save Impulse_1.csv, Impulse_2.csv, Impulse_3.csv. A stronger impulse should change amplitude; for the linear model it should not change the natural frequency. Without a force record, tap levels are qualitative; do not claim numerical impulses. A tap mainly sets initial velocity. In contrast, Lab 4 uses a small pull-and-release without pushing: initial displacement is nonzero and initial velocity is nearly zero. Both then follow the same free-decay equation.

4. Calibrate the harmonic input

Add 1-2 magnetic dampers and remeasure total moving mass. With the motor OFF, obtain a short free-decay record with both spring ends held stationary. Analyze it in mode 1 to obtain damping c for this configuration. A damping value from the no-magnet case is not appropriate here.

With the GTA, temporarily attach the cart directly to the driving hook as described in the prior setup procedure; keep the opposing spring attached. At low motor speed record the driving motion and fit a sine in Graphical Analysis. Alternatively measure the endpoint travel with a ruler. Target about 2-3 cm peak-to-peak; record the actual value. The input peak amplitude is Y = peak-to-peak/2. Stop the motor before reconnecting the driven-side spring. Measure the endpoint travel again if the pin setting or motion changes.

5. Harmonic response sweep

Reconnect the cart between the fixed-side and driven-side springs. With the GTA, increase the supply gradually within the prior setup range 2-11 V. Use the digital tachometer to measure rpm; voltage is not a frequency measurement. For one cycle per revolution: $f = \text{rpm}/60$ [Hz] and $\omega = 2\pi f$ [rad/s].

At each setting allow the transient to decay (often 1-2 minutes), then record at least 5 settled cycles, preferably 10 or more. Aim for 3-5 frequencies below and 3-5 above the predicted resonance, without operating at the peak. Keep travel safe and both springs taut. If the response grows excessively, stop and ask the GTA; do not dwell at resonance.

For every run record: filename, m, k_1, k_2, damper count, rpm, input peak-to-peak travel, and start/end times of the settled interval. Save each motor setting as a separate CSV. If motor speed drifts or the spring becomes slack, correct the setup and repeat.

6. Analyze in MATLAB

Download the Lab 5 MATLAB bundle, unzip it, and keep the three analysis files together. Open Run_Lab5_Forced_Response.m and click Run. Helper functions are called automatically; do not run them individually.

Impulse/free decay: choose mode 1. Enter measured mass and spring stiffnesses; select the raw CSV or Excel files. Enter time/position column numbers (or [] for clear headers), then select a clean time interval after hand contact ends. Choose meters. The program estimates frequency and damping and exports a data/fit plot and spectrum.

Harmonic sweep: choose mode 2. Enter mass, springs and c from the matching damped configuration. Select the harmonic files. For each one enter a settled interval, units, measured rpm, and input peak-to-peak displacement in meters. The program fits a sinusoid at the measured drive frequency, calculates X/Y, and overlays the analytical amplitude predicted from your measured parameters. Its display phase is aligned to the output; this does not measure input/output phase.

For Excel select the worksheet containing the run. If the export contains an index column or several runs, explicitly enter the time and position column numbers, e.g. [2 3]. The program stops on time resets or large gaps; export a single continuous run. Review the fit against the measured points before accepting results.

The model used for comparison

For cart displacement x(t) [m], moving spring-end displacement y(t) [m], damping c [N·s/m] acting between cart and stationary track, and total moving mass m [kg]:

$m\ddot{x} + c\dot{x} + (k_1 + k_2)x = k_2 y(t)$

The driving spring force is $k_2(y - x)$; the fixed spring force is $-k_1 x$. For $y = Y\cos(\omega t)$, the steady response is $x = X\cos(\omega t - \varphi)$. Compare the measured peak-amplitude ratio with

$X/Y = k_2 / \sqrt{[(k_1 + k_2 - m\omega^2)^2 + (c\omega)^2]}$.

Plot $20\log_{10}(X/Y)$ against ω on a logarithmic horizontal axis. At low frequency the ratio approaches $k_2/(k_1 + k_2)$, which is generally not 1. Use sine-fit amplitudes rather than FFT peak heights. The phase φ requires a synchronized input record and is not a required experimental deliverable.

Before leaving

Show the GTA one impulse trace and spectrum, one harmonic trace with a settled fit, and the frequency-response plot with at least five measured points. This is a quick check; finish the wider sweep for the report. Confirm every group member has the raw files and the parameter log.

7. What to include in the combined Labs 4 + 5 report

Use the course report structure and combine the two experiments in one report. Keep free-response (Lab 4) and forced-response (Lab 5) results clearly labeled. Use only equations that support your analysis; define every variable at first use.

Impulse results: combine the three impulse time histories in one figure and their spectra in another. Label the measured frequencies. Explain amplitude changes, frequency consistency and any departures from a linear model.

Harmonic results: include a representative time record showing startup/transient and settled response; mark the interval used for the sine fit. Report the drive frequency and fitted X and measured Y. The code only analyzes the selected interval, so use the original full Vernier record for the startup illustration.

Frequency response: overlay measured X/Y points and the theoretical curve, using your measured m, k_1, k_2 and matching c. Include a compact table of frequencies, amplitudes and ratios. Discuss low-frequency behavior, resonance amplification and high-frequency reduction. Do not invent a point at resonance.

Link to Lab 4: compare natural frequency and damping obtained from free decay with the forced-response behavior. Distinguish the natural frequency f_n, damped transient frequency f_d, and imposed drive frequency f. A steady-state forced oscillation follows the drive frequency.

Model comparison: the supplied analytical transfer response and the pre-lab state-equation simulation describe the same linear system. Explain disagreements using your evidence: spring slack, rolling friction, motor-speed changes, altered input travel or unsettled data. A good fit alone does not prove the physical model.

Useful checks

Cannot find the export? Select CSV/XLSX/XLS in the file picker. Do not select the Vernier project file or a folder.
Fit misses the dots? Check worksheet, columns, units, rpm and time window. Verify the input frequency with the Vernier trace.
Gain appears doubled? Both X and Y must be peak amplitudes; Y is half the input peak-to-peak travel.
Response still grows/decays? Choose a later interval; the program flags a first-half/second-half amplitude change greater than 10%.
Wrong prediction? Check kilograms, both spring stiffnesses, and damper mass and c for the actual run.

Files to keep

Raw CSV/Excel files, parameter log, calibration record, Lab5_Results folder and the report. Export figures directly; avoid screenshots of MATLAB windows. Due date: follow Canvas and the course syllabus (combined Labs 4 + 5 report: Monday, November 23, 11:59 PM).

Source: MIE 402 prior-course Lab 5 manual, theory and experiment notes; procedure adapted for Fall 2026. Numerical examples in the lecture and pre-lab are illustrative, not experimental results.